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In Collaboration with

ROBERT GORDON UNIVERSITY ABERDEEN

Nutri Scanner

(Personalized Nutrition Tracking and Health Risk Prediction System)

Group 10 Project Proposal Document by:

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1. INTRODUCTION

1.1. Chapter Overview

This project aims to introduce a sophisticated AI-powered nutrition tracking and health risk prediction system tailored specifically for the Sri Lankan context. Many individuals in Sri Lanka face challenges in managing dietary health due to the complexity of multilingual food labels and the lack of nutritional data for traditional homemade meals. Given the rising prevalence of non-communicable diseases such as diabetes, obesity, and hypertension, there is a critical need for accessible, localized dietary management tools.

The proposed solution is an automated mobile application that leverages **Computer Vision**, **Optical Character Recognition (OCR)**, and **Natural Language Processing (NLP)** to analyze food products and meal images. By integrating these technologies, the system provides personalized nutritional advice and predicts potential health risks based on individual user profiles. This project addresses the limitations of existing global health apps by supporting local languages and unique dietary patterns, ultimately empowering users with accurate, AI-driven insights to improve their long-term health outcomes.

1.2. Problem Domain

The nutritional landscape in Sri Lanka is characterized by a rich diversity of traditional home cooking and prepared foods; however, this variety is accompanied by a burgeoning public health crisis. Diet-related chronic conditions, including diabetes, hypertension, and heart disease, are on an epidemic rise, exacerbated by a significant gap in nutritional literacy. A study on "Nutritional labeling and its impact" highlights that while labeling is a crucial tool for informed decision-making, it is poorly utilized by Sri Lankan consumers due to small print, technical jargon, and complex formatting (Maduwearachchi, Jemziya, & Rifath, 2024).

The primary challenges contributing to this health decline are rooted in the inadequacy of existing digital solutions and the unique socioeconomic factors of the region:

Incompatibility of Global Solutions The majority of nutrition monitoring applications currently available in the marketplace are engineered for Western markets. These platforms are optimized for English labels and Western dietary patterns, failing to accommodate Sri Lanka's multilingual food culture and traditional recipes. Consequently, these applications lack the cultural and linguistic nuance required to be effective or widely adopted in the local context.

Public Health Crisis and Nutritional Disparities Sri Lanka faces a dual burden of malnutrition. While many children suffer from stunted growth due to nutrient deficiencies, a growing segment of the adult population is experiencing a rise in obesity and type 2 diabetes. These trends highlight an urgent necessity for localized research and targeted interventions to address specific dietary failures.

Barriers to Nutritional Awareness Even when consumers are motivated to eat healthily, they face several systemic barriers:

> **Linguistic Complexity:** Food labels are often replicated in duplicate using small text and jargon that the average consumer cannot decipher.
>
> **Data Scarcity:** There is a lack of accessible, verified nutritional data for traditional Sri Lankan meals (e.g., specific curries and local starches).
>
> **Economic Constraints:** Access to professional nutrition counselling is often expensive and inaccessible to the general population, leaving many without personalized guidance.

Educational Gaps: A general lack of awareness regarding the long-term health consequences of traditional high-carbohydrate or high-sodium diets.

The consequences of maintaining the status quo are severe, potentially leading to a permanent decline in national health standards and increased mortality from preventable non-communicable diseases. To mitigate this, a solution is required that closely monitors dietary intake and provides individualized, localized nutritional counselling.

1.3. Problem Definition

In Sri Lanka, most of the population cannot make informed food choices due to deceptive and unreadable food packaging. Processed food tends to have details both in Sinhala as well as English, but the script is very small and filled with technical terms that most consumers are unable to read (Perera, Jayawardena & Arunatilake, 2022; Wickramasinghe, Jayawardena & Arunatilake, 2025)

Even more so is the case with home-cooked foods, which tend not to have any details on the nutritional value (Perera, Jayawardena & Arunatilake, 2022). With no proper information, it is challenging to know the nutritional value of what they consume, and therefore, they find it even harder to eat healthily. This is further compounded by the reality that there are not many apps devoted to non-Western diets. They mostly employ international food databases and are available only in English, failing to capture Sri Lanka's extensive range of ingredients, traditional recipes, and food traditions. Furthermore, these devices simply provide generic nutritional information, without giving custom advice based on age, health status, or individual dietary needs.

Therefore, consumers face significant barriers to realizing the effect their food has on their well-being. Poor available, relevant, and understandable nutritional information impedes their choice towards healthier options, which creates an epidemic of diet-related illnesses across the country (Wijesundara, 2021; Perera, Jayawardena & Arunatilake, 2022).

1.4. Research Motivation

This research is motivated by the need to address the lack of localized nutritional analysis in Sri Lanka, a leading factor in dietary mismanagement that is often disregarded by existing global health applications. By developing an enhanced recognition system of both packaged and traditional foods through advanced computer vision and integrated datasets, we aim to significantly bridge the regional data gap. The goal is to empower Sri Lankan users to navigate their unique dietary landscape more effectively, providing accurate nutritional insights and timely health risk predictions. By making localized information accessible, this system aims to promote healthier eating habits and offers a sustainable solution for long-term health management, ultimately improving the overall quality of life within the local community

1.5. Existing Work

Table 1: Existing Work

Citation	Technology / Algorithm Used	Advantage	Limitation
(Patil, Londhe & Timmapuri, 2025)	CNN-based OCR for nutrition info extraction	Achieved ~75% text extraction accuracy even with low-quality labels; strong barcode recognition	Struggles with complex multilingual or non-standard packaging

(Lodha et al., 2025)	Multi-stage OCR +CNN classification	High ingredient classification accuracy (92.4%); robust for diverse layouts	Limited applicability to local (Sri Lankan) label structures
(Hettiarachchi et al., 2020)	OCR+Text preprocessing on bilingual labels	Highlighted challenges and potential solutions for Sinhala-English packaging	Lacks full automation; dataset limited to experimental samples
(Hen et al., 2023)	Depth prediction CNN fusion model	<10% mean absolute error for calorie prediction	Dataset-specific model;imited generalization to new foods
(Zhou et al., 2023)	Knowledge graph Embedding model	Improved normalization/linking accuracy by 12%	Requires well-structured ontology; high computational complexity
(University of Moratuwa, 2021)	RNN LSTM Ontology-based diet system	Culturally tailored meal recommendations for Sri Lankan context	Focused on meal planning, not label interpretation

1.6. Research gap

Existing Datasets for both images of Sri Lankan foods as well as in-depth nutritional analysis is lacking. Along with experts in the field, we hope to rectify this issue by filling this gap through collecting our own images as well as creating a composite dataset verified by credited researchers. This will ensure that our system fulfills its goal to create a system catered specifically to the health-conscious Sri Lankan.

Existing Food scanning and nutrition scanning Systems and applications that are already available on the market have low accuracies and a lack of data to prove their feasibility. As stated, before similar systems do not have adequate data to provide reliable information about Sri Lankan foods and their nutrition.

1.7. Contribution of the body knowledge

1.7.1. Domain Contribution

The Nutri Scanner provides a localized, AI-driven solution for recognizing Sri Lankan foods, extracting nutritional data, and predicting health risks. Key contributions include:

1. **Localized Nutritional Intelligence**: Unlike Western-centric apps, this system is specifically trained to interpret Sri Lankan food labels and traditional meals, ensuring relevance for local consumers.
2. **Hybrid Analysis of Packaged and Homemade Foods**: By combining OCR for label reading with computer vision for plate recognition, the system tracks both processed products and unlabeled homemade meals for a comprehensive dietary overview.

3. **Predictive Health Analytics**: The system goes beyond tracking by using machine learning to link nutrient intake to risks for diabetes, hypertension, and obesity, enabling early dietary intervention and personalized health management

.1.7.2. Technological Contribution

he Nutri Scanner project is a unified nutrition intelligence platform that advances applied AI research by developing localized datasets and models specifically for Sri Lankan food labels and traditional dishes. It introduces an AI framework for tracking nutrition in low-resource settings through the integration of computer vision and OCR for bilingual label analysis. Key technological contributions include AI-based nutrient extraction and normalization, machine learning-driven health risk prediction, and personalized recommendations tailored to local dietary patterns.

1.8. Research Challenges

- **Heterogeneous Data Integration:** Merging diverse sources—OCR label data, image recognition outputs, and user health profiles—into a unified structure is complex. Precise alignment and normalization are critical for accurate nutrient estimation and risk prediction.

- **Dataset Scarcity:** A lack of public datasets for Sri Lankan packaged foods and traditional meals necessitates extensive manual collection. Limited diversity in training data risks reducing model generalization and accuracy.

- **OCR Obstacles:** Small fonts, curved packaging, and reflective surfaces on food labels distort text. Ensuring robust extraction of nutritional values under these varying conditions is a significant hurdle.

- **Visual Variability in Meals:** Sri Lankan dishes vary by cooking style and ingredients. Overlapping items and mixed textures complicate image segmentation, making accurate identification difficult.

- **Portion Size Estimation:** Calculating volume from a single image without physical reference objects is technically challenging. Camera angles and lighting variations often lead to inaccurate nutritional calculations.

- **Risk Prediction Reliability:** Predictive accuracy depends on the precision of earlier stages. Errors in food recognition or nutrient extraction propagate through the system, potentially compromising the reliability of health recommendations.

- **Computational Efficiency:** Executing vision, OCR, and predictive models in real-time on mobile devices strains resources. Balancing processing speed with model accuracy is essential for a responsive user experience.

1.9. research questions

1. The study investigates the effective combination of food label OCR technology and food image recognition results to generate precise nutritional intake measurements.

2. The study evaluates how well extracted nutritional information enables daily nutrient estimation compared to manual food logging.

3. The system achieves its goal of identifying packaged and homemade foods through various image recognition methods which help reduce recognition errors.

4. The study will use specific assessment methods to evaluate the performance of computer vision systems and prediction models which will determine their ability to estimate nutrients and predict health risks.

1.10. Research Aim

To conclude, the research's aim is to attain a system that is capable of identifying an array of Sri Lankan Foods and provide users with accurate and easy to assess information, it also seeks to rectify the gap of Sri Lankan food knowledge bases through collaboration and individual data collection.

1.11. Research Objective

Table 2: Research Objectives

Research Objective	Explanation	Learning Outcome
Problem Identification	**RO1: Data Extraction & Recognition** The project successfully develops a system capable of extracting and recognizing nutritional information from labels on packaged foods, addressing the language barriers Sri Lankan consumers face when interpreting dietary information. **RO2: Data Normalization and Imputation** The system demonstrates effective data normalization and imputation techniques to standardize nutritional information from diverse sources, ensuring consistent and comparable dietary data across different food products and databases. **RO3: Health Risk Prediction Modeling** The system achieves accurate health risk prediction by analyzing user input data (age, gender, weight, activity level, medical conditions) combined with nutritional intake patterns, providing personalized health risk assessments. **RO4: Contextualized Recommendation Generation** The project delivers contextualized dietary recommendations tailored to individual user profiles, taking into account personal health data, existing medical conditions, and cultural food preferences relevant to Sri Lankan consumers. **RO5: End-to-End System Integration and Feedback** The system is designed with complete integration from food image capture through OCR/Computer Vision analysis to nutritional insights and recommendations, with continuous user feedback mechanisms for system improvement and validation. **RO6: Computer Vision and OCR for Food Detection** The inclusion of Computer Vision and OCR technologies successfully enables automated analysis of nutrition labels and homemade meals, providing personalized, risk-based nutritional insights at scale. **RO7: Scalable Nutrition Estimation** The project demonstrates that accurate nutrition estimation can be achieved by integrating reliable data extraction methods with principled normalization techniques and canonical databases (such as USDA FoodData Central and Sri Lankan food composition tables), creating a cost-effective and accessible solution for dietary tracking.	LO1

Literature Review	The literature review aims to cover already explored research that has been carried out on nutritional label analysis, food recognition, health risk prediction, and dietary recommendation systems. The review will identify similar works, cite useful documents and articles that prove certain facts and claims within the domain of nutrition tracking and health monitoring for diverse populations, and address ethical constraints regarding data privacy and personalized health information.	LO1
Data Gathering and analysis	• Interviews with nutritionists and healthcare professionals to understand dietary needs and challenges faced by Sri Lankan consumers • Questionnaires on system usefulness and acceptance among local users • Data for research papers collected from IEEE DataPort, Google Scholar, etc. • Journal, Articles, Books published within 2021 to 2024	LO2,LO3
Research Design	Quasi-experimental design – since randomization is not possible due to ethical and logistical constraints, the study compares outcomes to evaluate the effectiveness of the nutrition analysis system in predicting health risks and improving dietary awareness among users.	LO3, LO4
Implementation	1**. Implementation of Computer Vision and OCR,** nutrition detection to analyze multilingual labels and homemade meals. 2. **Development of data extraction, normalization**, nutrient estimation with canonical databases and predictive models 3. **Implementation of nutritional analysis and interpretation** systems that convert raw nutrient data into actionable insights regarding, nutrient adequacy, and potential health outcomes 4**. Development of health risk prediction and personalized** recommendation systems that analyze user input data and nutritional levels to predict health risks and generate suitable diet plans on daily and weekly basis	LO2, LO3, LO4
Testing and Evaluation	To evaluate system performance using accuracy, F1-score, recall, and user feedback, ensuring reliability and usability for local users.	LO2, LO4

1.12. Project Scope

1.12.1. In-scope

1. Interpret nutritional data on Sri Lankan traditional food
2. Interpret nutritional data on Sri Lankan packaged products
3. Category based identification on broader items
4. OCR and CV as input.

1.12.2.Out-Scope

1. In-depth datasets on food found outside of Sri Lanka
2. Multilingual Label Translation beyond Sinhala and English
3. Complete analytical accuracy
4. Interpret most regional dishes beyond Sri Lanka

1.12.3.Prototype Diagram

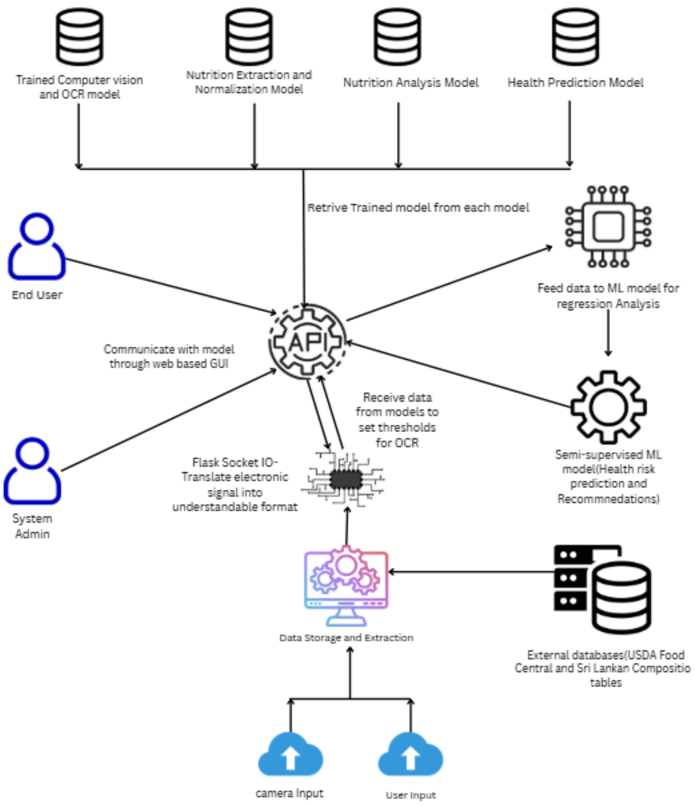

Figure 1: Prototype diagram

1.12.4 Process Breakdown

1. **Data Input and Capture:** Users upload food images or enter personal health data through the interface, and all inputs are sent to the system via an API.

2. **Image Processing and Classification:** Images are processed using OpenCV to generate analyzable numerical data and are handled in real time through Flask-SocketIO.

3. **Data Extraction and Recognition:** OCR extracts multilingual nutritional information from packaged foods, while computer vision identifies food items and Sri Lankan dishes from meal images. Extracted data is normalized and validated using USDA FoodData Central and Sri Lankan food composition tables.

4. **Model Retrieval and Processing:** Trained models for vision, OCR, nutrition analysis, and health prediction are retrieved through the Django REST API to process the extracted data.

5. **Data Integration and Analysis:** Nutritional data is combined with user profiles and analyzed using regression and semi-supervised machine learning models to assess health risks, nutrient adequacy, and personalized recommendations.

6. **Results Delivery and Alerts:** Results are displayed through the web interface, while important health alerts and goal achievements are communicated via SMS or email using the Twilio API.

7. **System Management and Feedback:** System administrators monitor performance and errors, and user feedback is collected to improve system reliability and model accuracy.

1.13 Resource Requirements

1. 13.1.Hardware Requirements

a. CPU & GPU: Intel Core i7 10th Generation (or equivalent) Nvidia 4060 TI 8GB - required to efficiently handle image processing, machine learning model training, and data analysis tasks.
b. Memory (RAM): 32 GB RAM - required for training and handling large datasets.
c. Storage: Minimum 512 GB SSD - for storing datasets, trained models, and project files.
d. Internet Connectivity: Required for accessing cloud platforms, APIs, dataset downloads, and collaborative development tools.

1.13.2. Software Requirements

a. Operating System: Windows / macOS
b. Programming Language: Python
c. Integrated Development Environment (IDE): PyCharm and Google Colab
d. Machine Learning Frameworks: TensorFlow and PyTorch
e. Computer Vision & Data Processing Libraries: OpenCV, NumPy, Pandas
f. Dataset Management Tool: Roboflow (for image labeling, augmentation, and dataset versioning)
g. Database Systems: MySQL or SQLite for storing user and nutrition-related data
h. Frontend Technologies: HTML, CSS, and JavaScript
i. Backend Technologies: PHP or Node.js
j. Documentation & Analysis Tools: Microsoft Word, R for statistical analysis

1.13.3 Data Requirements

• Global nutritional datasets for standardized nutrient values
• Sri Lankan food datasets to support local dietary patterns
• Food label data
• Images of Sri Lankan plated meals and packaged food items
• Custom-built dataset created using Roboflow for food classification
• Simulated user health profiles for testing health risk prediction models

1.13.4. Skill Requirements

a. Knowledge of machine learning and computer vision
b. Python programming and data preprocessing skills
c. Experience with dataset creation and annotation
d. Understanding of nutritional analysis concepts
e. Problem-solving and analytical thinking
f. Report writing and technical documentation skills

1.14 Chapter Summary

This chapter provided an overview of the Nutri Scanner project by discussing the problem domain, research motivation, objectives, scope, and resource requirements. It highlighted the need for an AI-based nutrition analysis system tailored to Sri Lankan dietary practices and outlined the tools, datasets, and methodologies required for its development. The chapter establishes a foundation for the subsequent chapters by defining the project direction and technical feasibility.

1.15 References

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